a

1 Locate

Elaborate the unchanged source; find the unique `TacticInfo` node of the authored tactic t_q^{auth}.

Frozen target q

→

2 Recover

Read $s_q = (C, M, G)$ from the node. Replay t_q^{auth}; its endpoint must equal $\kappa(s_q^+)$.

Validated state s_q

→

3 Execute

Run each $t_i \in L_q$ from a fresh copy of s_q; record its status and, on success, $\kappa(s_i')$.

23 branches per root

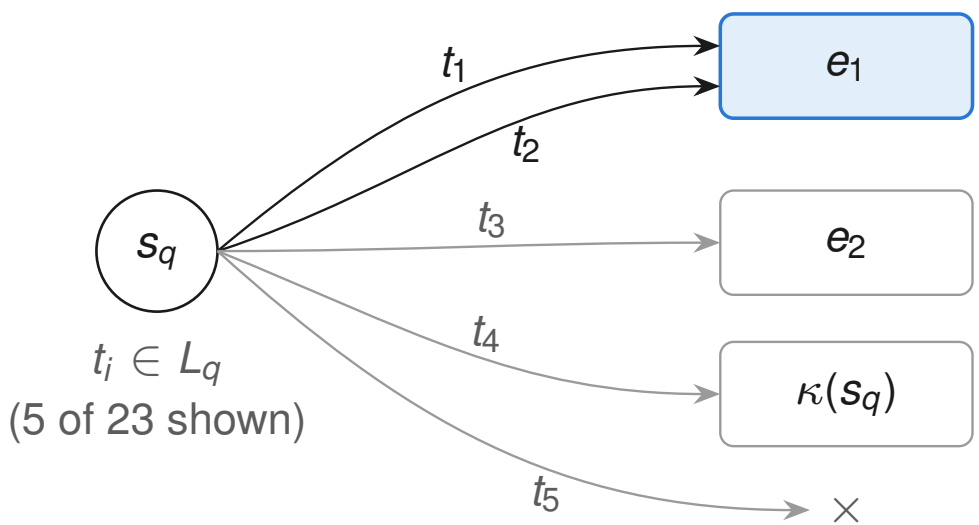
→

4 Compare

Group successful branches by κ. A witness is a pair with $h(t_i) \neq h(t_j)$ and $\kappa(s_i') = \kappa(s_j') \neq \kappa(s_q)$.

Root indicator W_q

b



$h(t_1) \neq h(t_2)$ and $\kappa(s_1') = \kappa(s_2') = e_1 \neq \kappa(s_q)$: a witness, so $W_q = 1$

A successful branch whose endpoint no other branch reaches

Endpoint equal to the start; excluded from witnesses

Unsuccessful attempt; no endpoint